

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 7

Groups: The Language of Symmetry

Exercises

Self-Study Curriculum

1. The integers under addition form a group. Verify each axiom. What is the identity? What is the inverse of 7? Now consider the integers under multiplication. Do they form a group? If not, which axiom fails?

2. The set $\{0, 1, 2, 3\}$ with addition modulo 4 forms a group ($\mathbb{Z}_4$). Write out the full multiplication table. Find all subgroups. For each subgroup, describe the corresponding quotient group. What is the quotient “collapsing”?

3. In data science, a “transformation” of a dataset (standardization, normalization, log-transform) can be composed, some can be inverted, and the identity transformation exists. Under what conditions do the transformations you commonly use form a group? When do they fail to? What does this tell you about the limits of “undoing” data preprocessing?

4. Two molecules have different chemical compositions but the same symmetry group. A spectroscopist says they will have the same number of distinct vibrational frequencies. Why? What

property of group representations makes this prediction possible?
